

Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

Ankit Gupta, Guy Dar, Shaya Goodman, David Ciprut, and Jonathan Berant. Memory-efficient transformers via top- k attention. *arXiv preprint arXiv:2106.06899*, 2021.

Appendix

A Question-Answering Granular Experiment Results

Table 4: GSM8K and MedQA Question Answering BERTScore

		GSM8K			Medqa		
Cache Budget / Fastgen Attn Recovery Frac (%)	Strategy	Precision	Recall	F1	Precision	Recall	F1
10	HASHEVICT	0.859	0.806	0.831	0.857	0.808	0.832
	L_2	0.858	0.798	0.826	0.833	0.813	0.823
	H ₂ O	0.877	0.830	0.853	0.866	0.795	0.829
	Scissorhands	0.873	0.825	0.848	0.867	0.795	0.829
30	HASHEVICT	0.893	0.854	0.873	0.867	0.834	0.850
	L_2	0.885	0.847	0.865	0.855	0.834	0.844
	H ₂ O	0.893	0.860	0.877	0.878	0.802	0.838
	Scissorhands	0.893	0.858	0.875	0.877	0.802	0.838
50	HASHEVICT	0.897	0.865	0.880	0.869	0.842	0.855
	L_2	0.891	0.861	0.875	0.866	0.841	0.853
	H ₂ O	0.896	0.866	0.881	0.879	0.803	0.839
	Scissorhands	0.896	0.864	0.879	0.878	0.804	0.839
70	HASHEVICT	0.896	0.866	0.881	0.869	0.843	0.855
	L_2	0.894	0.865	0.879	0.868	0.842	0.855
	H ₂ O	0.897	0.867	0.881	0.879	0.801	0.838
	Scissorhands	0.896	0.864	0.880	0.879	0.803	0.839
90	HASHEVICT	0.897	0.867	0.881	0.868	0.843	0.855
	L_2	0.896	0.866	0.881	0.868	0.843	0.855
	H ₂ O	0.897	0.867	0.881	0.879	0.801	0.838
	Scissorhands	0.896	0.864	0.880	0.880	0.802	0.839
50	Fastgen	0.811	0.770	0.789	0.816	0.763	0.788
60		0.827	0.778	0.801	0.806	0.766	0.785
70		0.837	0.788	0.811	0.811	0.766	0.787
80		0.874	0.840	0.857	0.866	0.793	0.828
90		0.896	0.864	0.879	0.876	0.800	0.836
100	Full	0.897	0.867	0.882	0.868	0.843	0.855

B Results of ablation on Hash Dimension

Please see Table 10

C Attention Scores and Key Norms Visualization

We further examine the method of our chief competitor, the L_2 eviction method [Devoto et al., 2024]. In particular, in Figure 7 we examine the key-norm-attention correlation suggested by the authors. Indeed, low key-norms, even across prompts, demonstrate a strong correlation with attention score.

D Attention Loss Ratio Analysis

We perform an attention loss ratio (ALR) analysis between LSH-based ranking and L_2 -based ranking. Our implementation is an adaptation of the methodology described in Devoto et al. [2024]. This section explores how much of the uncompressed attention matrix is preserved between HASHEVICT and the L_2 eviction strategy in Devoto et al. [2024].

Table 5: GSM8K Question Answering Rouge

Cache Budget / Fastgen Attn Recovery Fra(%)	Strategy	Rouge 1	Rouge 2	Rouge L	Rouge Lsum
10	HAShEViCT	0.206	0.051	0.157	0.186
	L_2	0.196	0.050	0.151	0.179
	H ₂ O	0.300	0.090	0.227	0.263
	Scissorhands	0.271	0.074	0.205	0.238
30	HAShEViCT	0.446	0.187	0.341	0.383
	L_2	0.392	0.149	0.288	0.337
	H ₂ O	0.481	0.208	0.364	0.410
	Scissorhands	0.471	0.203	0.357	0.403
50	HAShEViCT	0.511	0.234	0.393	0.438
	L_2	0.476	0.205	0.355	0.409
	H ₂ O	0.517	0.238	0.398	0.442
	Scissorhands	0.509	0.232	0.389	0.433
70	HAShEViCT	0.521	0.240	0.401	0.446
	L_2	0.509	0.230	0.386	0.435
	H ₂ O	0.523	0.243	0.404	0.446
	Scissorhands	0.510	0.233	0.392	0.435
90	HAShEViCT	0.525	0.243	0.403	0.449
	L_2	0.522	0.241	0.400	0.446
	H ₂ O	0.523	0.243	0.406	0.446
	Scissorhands	0.512	0.235	0.393	0.436
50	Fastgen	0.112	0.017	0.095	0.106
60		0.133	0.024	0.113	0.126
70		0.171	0.036	0.139	0.160
80		0.356	0.128	0.264	0.305
90		0.509	0.231	0.391	0.434
100	Full	0.526	0.244	0.403	0.449

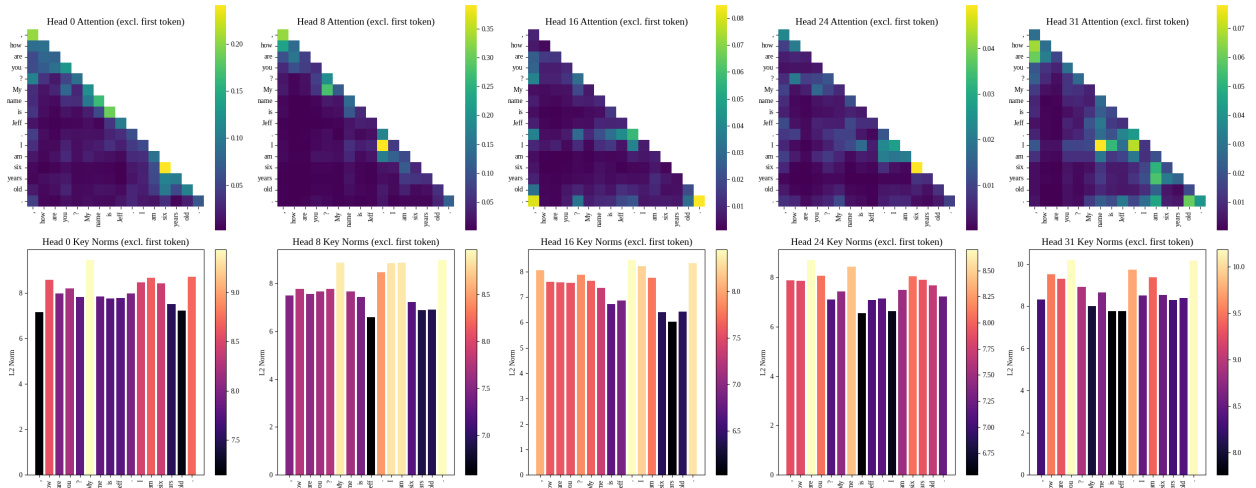


Figure 7: **Attention and Key Norms.** Attention scores and corresponding L_2 norms of key vectors (excluding the first token) for a sample of heads (0,8,16,24,31) in the 8th layer for a sample input sequence. Each subplot shows the attention heatmap (top) and the corresponding key norm values (bottom) for a particular head, allowing for a direct comparison between attention patterns and key norm values across different heads.